

Rounded polygons: definitions, perimeter, area, and gradients

1 Definitions

A rounded polygon is a CCW backbone polygon with vertices $\vec{v}_i$ ($i = 1, \dots, n$) whose corners are replaced by circular arcs of radius ρ . Let $z'(i)$ and $z(i)$ be the previous and next vertices of i . Define the *unit* incoming and outgoing edge vectors and their lengths

$$\hat{e}_{z'(i)} = \frac{\vec{v}_i - \vec{v}_{z'(i)}}{\ell_{z'(i)}}, \quad \hat{e}_i = \frac{\vec{v}_{z(i)} - \vec{v}_i}{\ell_i}, \quad \ell_{z'(i)} = |\vec{v}_i - \vec{v}_{z'(i)}|, \quad \ell_i = |\vec{v}_{z(i)} - \vec{v}_i|.$$

The corner (wedge) angle, kiss offset, kiss points, bisector, center, and swept angle are

$$\begin{aligned} \theta_i &= \arccos(-\hat{e}_{z'(i)} \cdot \hat{e}_i) \in [0, \pi], & t_i &= \frac{\rho}{\tan(\theta_i/2)}, \\ \vec{a}^- &= \vec{v}_i - t_i \hat{e}_{z'(i)}, & \vec{a}^+ &= \vec{v}_i + t_i \hat{e}_i, & \hat{b} &= \frac{\hat{e}_i - \hat{e}_{z'(i)}}{|\hat{e}_i - \hat{e}_{z'(i)}|}, \\ \vec{z} &= \vec{v}_i + \frac{\rho}{\sin(\theta_i/2)} \hat{b}, & \psi_i &= \pi - \theta_i. \end{aligned}$$

2 Perimeter

Each corner trades two straight pieces of length t_i for an arc of length $\rho\psi_i$, so $\Delta p_i = -2t_i + \rho\psi_i$ and

$$P = P_{\text{backbone}} - \sum_{i=1}^n (2t_i - \rho\psi_i).$$

Gradient. The backbone term contributes $\hat{e}_{z'(k)} - \hat{e}_k$ (the edge unit vectors meeting at k). With $\partial t_i / \partial \theta_i = -\rho / (2 \sin^2(\theta_i/2))$ and $\psi_i = \pi - \theta_i$, the per-corner coefficient is $-2 \partial t_i / \partial \theta_i + \rho \partial \psi_i / \partial \theta_i = \rho / \sin^2(\theta_i/2) - \rho = \rho \cot^2(\theta_i/2)$, hence

$$\boxed{\frac{\partial P}{\partial v_k^\alpha} = \hat{e}_{z'(k)}^\alpha - \hat{e}_k^\alpha + \rho \sum_{i=1}^n \cot^2\left(\frac{\theta_i}{2}\right) \frac{\partial \theta_i}{\partial v_k^\alpha}.$$

3 Area

Cut each corner with the chord $\vec{a}^- \rightarrow \vec{a}^+$ and split the corner correction into a *kite* minus a *wedge* (circular sector):

$$A = A_{\text{backbone}} - \sum_{i=1}^n s_i (A_i^{\text{kite}} - A_i^{\text{wedge}}), \quad s_i = \begin{cases} +1 & \text{convex} \\ -1 & \text{reflex.} \end{cases}$$

The kite $(\vec{v}_i, \vec{a}^-, \vec{z}, \vec{a}^+)$ is symmetric, so its perpendicular diagonals $|\vec{a}^+ - \vec{a}^-| = 2t_i \sin(\theta_i/2)$ and $|\vec{v}_i - \vec{z}| = \rho / \sin(\theta_i/2)$ give

$$A_i^{\text{kite}} = \frac{1}{2} (2t_i \sin \frac{\theta_i}{2}) \frac{\rho}{\sin(\theta_i/2)} = \rho t_i, \quad A_i^{\text{wedge}} = \frac{1}{2} \rho^2 \psi_i.$$

(Equivalently $A_i^{\text{kite}} - A_i^{\text{wedge}} = \frac{1}{2} t_i^2 \sin \theta_i - \frac{1}{2} \rho^2 (\psi_i - \sin \psi_i)$, i.e. corner triangle minus circular segment.)

Gradient. The correction $A_i^{\text{kite}} - A_i^{\text{wedge}} = \rho t_i - \frac{1}{2} \rho^2 \psi_i$ depends on the vertices only through θ_i , with

$$\frac{\partial}{\partial \theta_i} \left(\rho t_i - \frac{1}{2} \rho^2 \psi_i \right) = \rho \frac{\partial t_i}{\partial \theta_i} + \frac{1}{2} \rho^2 = -\frac{1}{2} \rho^2 \cot^2 \left(\frac{\theta_i}{2} \right).$$

Together with the backbone shoelace gradient $\partial A_{\text{backbone}} / \partial v_k^\alpha = \frac{1}{2} \varepsilon^\alpha_\beta (\vec{v}_{z(k)} - \vec{v}_{z'(k)})^\beta$ (where $\varepsilon^x_y = 1 = -\varepsilon^y_x$, giving $\frac{1}{2} (y_{z(k)} - y_{z'(k)}, x_{z'(k)} - x_{z(k)})$),

$$\boxed{\frac{\partial A}{\partial v_k^\alpha} = \frac{1}{2} \varepsilon^\alpha_\beta (\vec{v}_{z(k)} - \vec{v}_{z'(k)})^\beta + \frac{1}{2} \rho^2 \sum_{i=1}^n s_i \cot^2 \left(\frac{\theta_i}{2} \right) \frac{\partial \theta_i}{\partial v_k^\alpha}.}$$

The corner sums for ∂P and ∂A share the factor $\cot^2(\theta_i/2) \partial \theta_i / \partial v_k$, differing only by ρ vs. $\frac{1}{2} \rho^2 s_i$.

4 The angle gradient $\partial \theta / \partial V$

Differentiating $\theta_k = \arccos(-\hat{e}_{z'(k)} \cdot \hat{e}_k)$ and using the unit-vector derivative $\partial \hat{e} / \partial V = (\mathbb{I} - \hat{e} \hat{e}^\top) / \ell$ (this $1/\ell$ normalization is essential— $\partial \theta / \partial V$ has units of inverse length), only $j \in \{z'(k), k, z(k)\}$ contribute and

$$\boxed{\frac{\partial \theta_k}{\partial v_j^\alpha} = \frac{1}{\sin \theta_k} \left[\frac{\delta_{kj} - \delta_{z'(k)j}}{\ell_{z'(k)}} \left(\hat{e}_k^\alpha - (\hat{e}_k \cdot \hat{e}_{z'(k)}) \hat{e}_{z'(k)}^\alpha \right) + \frac{\delta_{z(k)j} - \delta_{kj}}{\ell_k} \left(\hat{e}_{z'(k)}^\alpha - (\hat{e}_k \cdot \hat{e}_{z'(k)}) \hat{e}_k^\alpha \right) \right].}$$

Writing $\hat{u} = (\vec{v}_{z'(k)} - \vec{v}_k) / \ell_{z'(k)} = -\hat{e}_{z'(k)}$ (toward prev) and $\hat{w} = (\vec{v}_{z(k)} - \vec{v}_k) / \ell_k = \hat{e}_k$ (toward next), so $\hat{e}_k \cdot \hat{e}_{z'(k)} = -\cos \theta_k$, the three contributions read

$$\frac{\partial \theta_k}{\partial \vec{v}_{z'(k)}} = -\frac{\hat{w} - \cos \theta_k \hat{u}}{\ell_{z'(k)} \sin \theta_k}, \quad \frac{\partial \theta_k}{\partial \vec{v}_{z(k)}} = -\frac{\hat{u} - \cos \theta_k \hat{w}}{\ell_k \sin \theta_k}, \quad \frac{\partial \theta_k}{\partial \vec{v}_k} = -\left(\frac{\partial \theta_k}{\partial \vec{v}_{z'(k)}} + \frac{\partial \theta_k}{\partial \vec{v}_{z(k)}} \right).$$

This is the routine `cornerAngleGradients`; the perimeter and area gradients above are `roundedPerimeterGradients` and `roundedAreaGradient`, all finite-difference checked to $\sim 10^{-10}$.